

OLSZTYN, Poland

WMO: 122720

Lat: 53.7686N

Lon: 20.4213E

Elev: 449

StdP: 14.46

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.0	8.1	-2.7	4.8	3.2	3.0	6.5	9.4	20.6	32.1	18.5	32.2	4.1	120	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	84.0	67.4	80.5	65.6	77.4	64.2	69.9	79.9	68.0	76.7	66.2	74.1	7.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	99.4	73.8	64.8	93.4	72.1	63.0	87.6	70.1	34.1	79.9	32.6	77.0	31.1	74.3	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.2	15.0	13.3	DB	-5.0	89.9	7.8	3.1	-10.6	92.2	-15.2	94.0	-19.6	95.8	-25.3	98.0	(4)
(5)			WB	-5.5	73.0	7.6	2.0	-11.0	74.5	-15.4	75.7	-19.7	76.8	-25.2	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.3	27.4	29.5	35.7	46.0	54.7	60.5	64.9	63.8	55.8	46.8	38.1	31.0	(6)
(7)		DBStd	15.12	9.99	9.44	7.83	7.80	7.42	6.15	5.67	5.25	5.84	6.82	7.11	9.20	(7)
(8)		HDD50	3029	700	575	445	166	35	1	0	0	16	145	358	588	(8)
(9)		HDD65	6998	1165	995	908	571	329	163	74	88	280	565	807	1053	(9)
(10)		CDD50	1668	0	0	2	46	181	316	461	427	190	44	1	0	(10)
(11)		CDD65	161	0	0	0	1	10	27	70	49	4	0	0	0	(11)
(12)		CDH74	1538	0	0	0	25	140	299	619	416	39	0	0	0	(12)
(13)		CDH80	349	0	0	0	3	16	71	172	83	4	0	0	0	(13)
(14)	Wind	WSAvg	6.5	7.3	7.2	7.4	6.6	6.5	6.3	5.9	5.4	5.5	6.4	6.9	7.1	(14)
(15)	Precipitation	PrecAvg	25.9	1.7	1.4	1.6	1.5	2.3	2.7	3.5	2.7	2.3	2.2	2.1	1.9	(15)
(16)		PrecMax	30.9	4.8	2.5	3.4	4.2	5.6	4.7	8.9	6.1	7.0	5.7	4.3	3.3	(16)
(17)		PrecMin	17.8	0.2	0.2	0.3	0.4	0.9	1.0	0.5	0.5	0.7	0.3	0.5	0.5	(17)
(18)		PrecStd	3.3	0.9	0.6	0.8	0.9	0.9	1.0	1.8	1.3	1.5	1.4	0.9	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.7	49.4	61.0	77.7	82.3	87.1	89.5	86.6	79.2	67.9	55.2	50.0	(19)
(20)			MCWB	44.6	45.0	49.5	59.6	63.9	68.9	69.6	70.2	64.3	58.8	52.1	48.1	(20)
(21)		2%	DB	44.1	46.6	55.3	70.5	78.2	81.4	84.8	82.0	73.7	63.5	51.7	46.8	(21)
(22)			MCWB	42.2	43.3	46.5	55.9	62.3	65.9	68.4	66.9	62.0	56.7	49.1	44.5	(22)
(23)		5%	DB	41.5	43.8	50.6	65.0	74.0	77.1	80.7	78.6	69.8	59.8	49.3	44.5	(23)
(24)			MCWB	39.8	41.3	44.7	52.3	60.1	63.2	66.5	65.1	59.8	55.1	47.1	42.8	(24)
(25)		10%	DB	39.0	41.2	46.9	60.3	69.5	73.5	76.8	75.3	66.4	56.8	47.2	41.9	(25)
(26)			MCWB	37.5	38.8	42.2	49.9	57.8	61.8	65.3	63.8	58.5	53.0	45.3	40.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.3	46.5	51.3	60.6	66.3	71.1	72.6	71.7	66.1	60.7	52.9	48.4	(27)
(28)			MCDWB	47.0	48.3	58.4	75.7	77.9	82.3	83.7	81.9	75.2	66.9	54.5	49.6	(28)
(29)		2%	WB	42.4	43.8	48.1	57.1	63.6	67.7	70.4	69.1	63.2	57.7	49.8	45.1	(29)
(30)			MCDWB	43.9	45.9	53.0	67.9	74.5	77.5	80.9	78.1	70.7	61.7	51.5	46.3	(30)
(31)		5%	WB	39.9	41.5	45.2	53.8	61.6	65.3	68.6	67.2	61.2	55.5	47.3	42.8	(31)
(32)			MCDWB	41.3	43.7	49.6	62.6	71.9	74.2	77.6	75.3	67.5	59.1	48.9	44.3	(32)
(33)		10%	WB	37.7	38.8	42.6	51.0	59.1	63.1	66.8	65.3	59.4	53.5	45.5	40.6	(33)
(34)			MCDWB	38.9	40.8	46.5	58.4	67.4	70.9	74.3	72.7	64.8	56.7	47.0	42.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	9.5	12.8	18.2	19.5	18.0	17.5	17.7	15.6	12.0	7.4	7.1	(35)
(36)			MCDBR	7.8	11.1	19.2	26.6	25.7	24.6	23.8	23.7	21.9	16.8	9.7	7.8	(36)
(37)			MCWBR	7.0	8.8	12.2	13.9	12.3	11.6	10.0	10.6	11.5	10.7	8.0	7.1	(37)
(38)		5% WB	MCDBR	7.8	10.2	17.2	23.8	23.0	21.5	20.7	19.7	18.6	15.1	9.0	7.7	(38)
(39)			MCWBR	7.0	8.8	12.0	13.4	12.1	11.6	10.0	9.9	10.5	10.2	7.9	7.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.302	0.332	0.365	0.410	0.399	0.398	0.430	0.420	0.377	0.357	0.343	0.301	(40)	
(41)		taud	2.228	2.237	2.248	2.190	2.281	2.313	2.241	2.270	2.368	2.412	2.310	2.263	(41)	
(42)		Ebn at Noon	205	238	256	258	269	270	258	253	251	229	188	181	(42)	
(43)		Edh at Noon	20	27	34	41	39	38	41	37	30	23	19	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	193	417	810	1310	1679	1749	1615	1389	987	517	195	130	(44)	
(45)		RadStd	34	70	129	171	162	155	178	131	129	88	23	15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	-256	+207	N/A
(47)	Regional (25 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	-263	+133	N/A

Nomenclature: See separate page